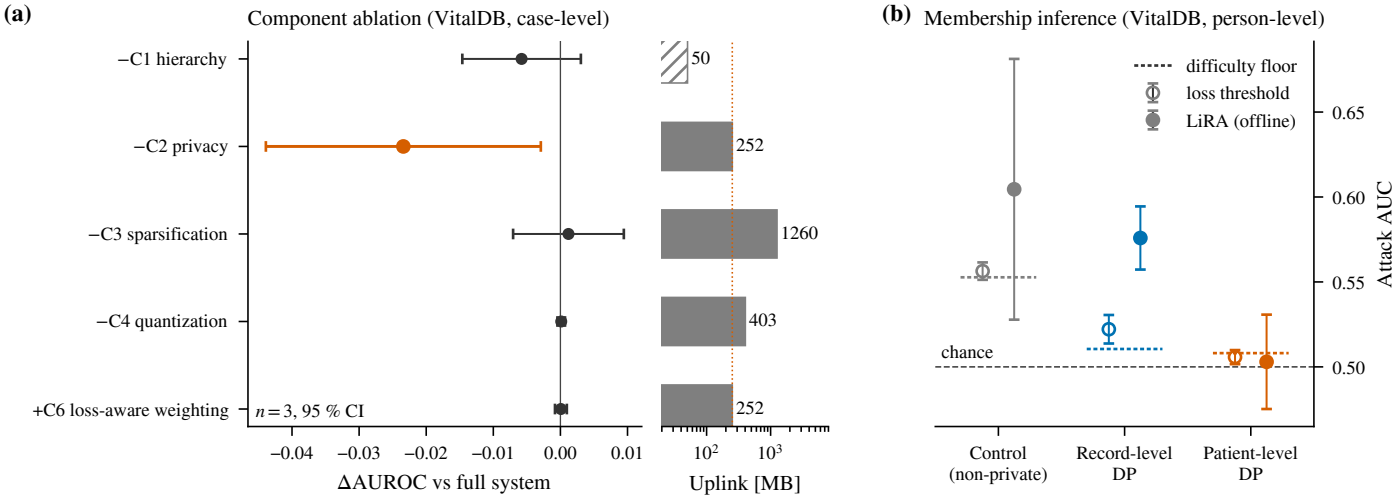
(a)
Component ablation (VitalDB, case-level)
−C1 hierarchy
−C2 privacy
−C3 sparsification
−C4 quantization
+C6 loss-aware weighting
n = 3, 95 % CI
−0.04
−0.03
−0.02
−0.01
0.00
0.01
ΔAUROC vs full system
50
252
1260
403
252
10²
10³
Uplink [MB]
(b)
Membership inference (VitalDB, person-level)
difficulty floor
loss threshold
LiRA (offline)
chance
0.65
0.60
0.55
0.50
Attack AUC
Control (non-private)
Record-level DP
Patient-level DP